PCG Arena

A Web Platform for Blind A/B Testing of Procedural Content Generators

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Artificial Intelligence ♡ Games:
Procedural Content Generation

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Outline

Outline

The PCG Evaluation Problem

Challenge: How do we know if a PCG algorithm produces "good" levels?

- Automated metrics miss player experience
- Lab studies have limited participants
- Generator reputation creates bias

Solution: Web-based blind A/B testing at scale

Research Questions

1. What makes levels "good"?
2. Are player preferences consistent?
3. Can we build a reward model?

PCG Arena: Core Concept

- Play two levels → Vote for favorite → Glicko-2 ranking
- **15 generators**, **748 levels**, **571 votes** collected

Outline

System Architecture

Frontend

- React + TypeScript
- Mario engine ported from Java
- Browser-based gameplay

Backend

- FastAPI (Python)
- Glicko-2 rating system
- AGIS matchmaking

Infrastructure

- Docker containers
- Google Cloud Platform
- SQLite database

Live at:

`https://pcg-arena.com`

Key Platform Features

For Players

- No download required
- Google OAuth login
- Tag gameplay characteristics

For Researchers

- Submit custom generators
- View live leaderboard
- Export research data

Why Glicko-2?

- Provides uncertainty estimates (RD)
- Handles inactive generators
- Maps pairwise comparisons to rankings

Win probability:

$$P(A \text{ wins}) = \frac{1}{1 + 10^{(R_B - R_A)/400}}$$

Parameter	Value
Initial Rating	1000
Initial RD	350
System τ	0.5

Table: Glicko-2 settings

Outline

14 PCG Algorithms Compared

Neural/ML-Based

- MarioGPT (GPT-2)
- MarioGAN (DCGAN)
- MarioDiffusion
- MarioDPO (ours)

Search-Based

- Genetic algorithm
- Hopper (agent simulation)

Grammar/Pattern-Based

- Notch variants (3)
- Pattern-based (3)
- ORE (occupancy)

Baseline

- **Original SMB** (15 levels)

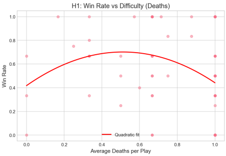
100 levels per generator (except Original: 15)

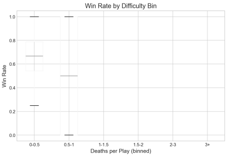
Outline

H1: Flow Channel Hypothesis

Hypothesis: Moderate difficulty (15–40% death rate) maximizes enjoyment.

Result: Not Supported – Linear relationship: easier = better ($r = -0.227$, $p = 0.046$)





Conclusion

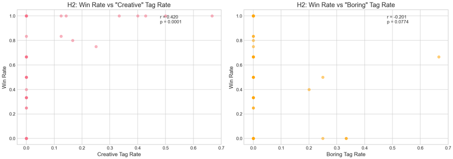
Players prefer levels they can complete. No inverted-U pattern found.

RQ1: Structural Variety

H2: Creativity Hypothesis

Hypothesis: Higher terrain variety and enemy diversity preferred.

Result: Partially Supported – “Creative” tag $\leftrightarrow$ wins ($r = 0.42$, $p < 0.01$)



Tag	r	p
Creative	0.416	0.003
Boring	-0.246	0.092

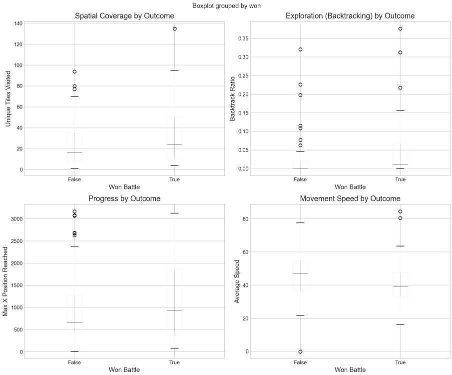
Conclusion

Levels tagged “creative” win significantly more often.

RQ1: Path Freedom & Exploration

H3: Agency Hypothesis

Hypothesis: Levels allowing more exploration are preferred.
Result: Partial Support – Backtrack ratio predicts wins ($p = 0.019$)



Metric	Won	Lost
Unique Tiles	68.7	42.1
Backtrack %	13%	21%
Max X	1533	1095

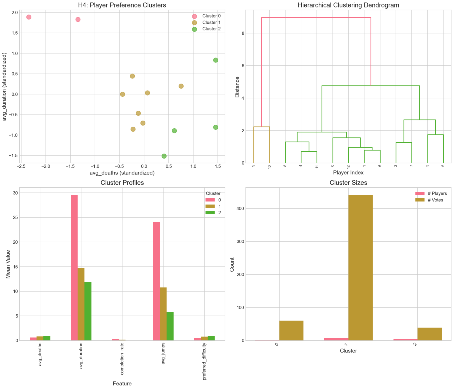
Conclusion

Winners explore 63% more tiles.
Non-linear paths preferred.

RQ2: Player Consistency

H4: Player Clusters Hypothesis

Hypothesis: Distinct player preference groups exist.
Result: Supported – 3 clusters identified via K-means



Player Types:

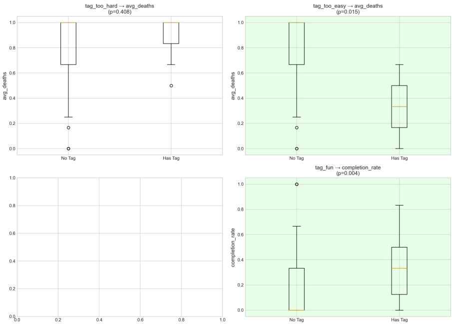
1. **Explorers** (n=2)
   Long sessions, high completion
2. **Mainstream** (n=6)
   Most votes, balanced play
3. **Strugglers** (n=4)
   Low completion, high deaths

RQ3: Tag Validation

H6: Tags Match Telemetry

Hypothesis: Subjective tags correlate with objective metrics.

Result: Supported – 5/6 tags match expected metric direction



Tag Semantics:

- fun ≈ playable + beatable
- creative ≈ engaging
- too_easy ≈ high completion
- too_hard ≈ high death rate

Conclusion

Tags can be trusted as quality signals for ML.

Outline

Building an Automated Judge Function

Goal: Create reward model for RLHF/DPO training

Stage 1 – Static Features:

$$J_{static} = \frac{w_{style}}{1 + D_M} - w_{gap} \cdot \text{Gap}$$

Stage 2 – Dynamic Features:

$$J_{final} = J_{static} + w_{vert} \cdot \sigma_y$$

Weight	Value	Source
w_{style}	0.32	Exp D
w_{vert}	0.26	Exp A
w_{gap}	0.50	Exp B

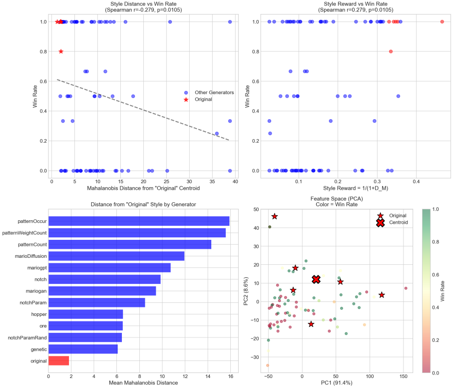
Key Result

Judge function correlates with human preference: **Spearman** $r = 0.812$, $p < 0.001$

Exp D: Mahalanobis Distance

Hypothesis: Levels closer to "Original" style win more.

Result: Supported – $r = -0.279$, $p = 0.011$



Conclusion

The "Nintendo Factor": original SMB levels define the quality target.

Style reward:

$$r_{style} = \frac{1}{1 + D_M}$$

where D_M = Mahalanobis distance

MarioDPO: Aligning Generators with Human Preferences

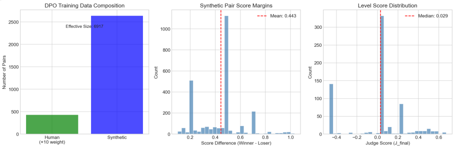
Approach: Direct Preference Optimization (DPO) fine-tuning

Training Data

- 571 human pairs ($\times 10$ weight)
- 3,500+ synthetic pairs (Judge)
- **9,200+ effective pairs**

Architecture

- Base: GPT-2 small
- Tokenization: Character-level
- DPO $\beta = 0.1$



MarioDPO Results: Beating Other PCG Methods

Rank	Generator	Rating	Games	W	L	T
1	Original Super Mario Bros (SMB1) Levels	1451	83	70	6	7
2	MarioDPO (Direct Preference Optimization)	1340	20	15	3	2
3	Occupancy-Regulated Extension (ORE) Generator	1221	91	55	25	11
4	Notch (Infinite Mario Bros) Generator	1158	77	44	23	10
5	Hopper Level Generator	1092	80	42	30	8
6	Grammatical Evolution (GE) Generator	1063	78	35	27	16
7	MarioGPT (Text2Level via GPT-2)	1045	88	49	32	7
8	MarioGAN (DCGAN + CMA-ES Latent-Space Generator)	1020	91	39	41	11
9	Pattern-based Generator (Weighted Pattern Count Fitness)	936	83	22	49	12
10	MarioDiffusion (Text-to-Level Diffusion)	936	84	26	40	18
11	Parameterized Notch (Randomized Params) Generator	905	88	27	45	16
12	Parameterized Notch Generator	853	82	16	49	17
13	Pattern-based Generator (Pattern Occurrence Fitness)	841	79	19	51	9
14	Pattern-based Generator (Pattern Count Fitness)	777	77	11	50	16

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Key Achievement

MarioDPO achieves highest Elo among all PCG generators, second only to hand-crafted Original SMB levels!

Generator Rankings Summary

Rank	Generator	Win Rate
1	**Original (SMB)**	89.6%
2	MarioDPO (ours)	83.3%
3	ORE	69.3%
4	Notch	67.4%
5	MarioGPT	63.8%
12	patternOccur	26.1%
13	notchParam	23.9%
14	patternCount	20.4%

Key Insights

- Original dominates (89%)
- MarioDPO beats other ML methods
- Pattern generators fail
- Style distance predicts quality

Outline

Key Findings

1. **Playability $>$ Challenge**
   Players prefer levels they can complete (no flow channel)
2. **Exploration Matters**
   Non-linear, explorable levels win 63% more tiles visited
3. **Style Matching Works**
   Closer to Original SMB $\rightarrow$ higher win rate ($r = 0.28$)
4. **Tags are Valid Signals**
   Subjective tags correlate with objective metrics (5/6)
5. **DPO Alignment Succeeds**
   MarioDPO achieves highest Elo among PCG generators

Contributions

Platform

- Open-source web platform for PCG evaluation
- Mario engine ported Java $\rightarrow$ TypeScript
- Glicko-2 + AGIS matchmaking

Research

- Empirical comparison of 15 generators
- Validated judge function ($r = 0.28$)
- MarioDPO: SOTA among PCG methods (83.3%)

Data

- 748 levels, 571 human votes
- Player trajectories and death heatmaps
- Open dataset for future research

Future Work

Short-term

- Collect more votes
- Add more generators
- Improve matchmaking

Long-term

- Transfer to other games
- Online DPO training
- Multi-objective PCG

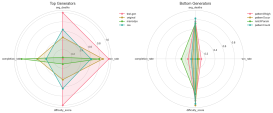
https://pcg-arena.com

Try it yourself!

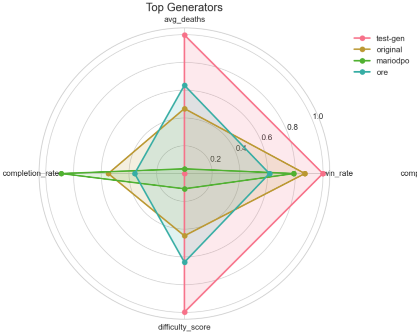
Thank You!

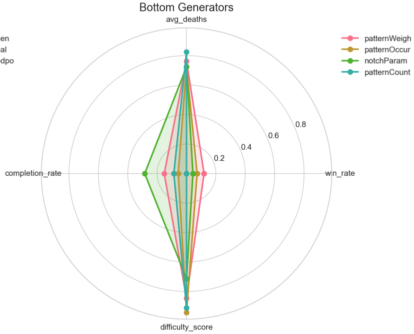
Questions?

Platform: `https://pcg-arena.com`
Code: `https://github.com/antoniczapski/pcg-arena`

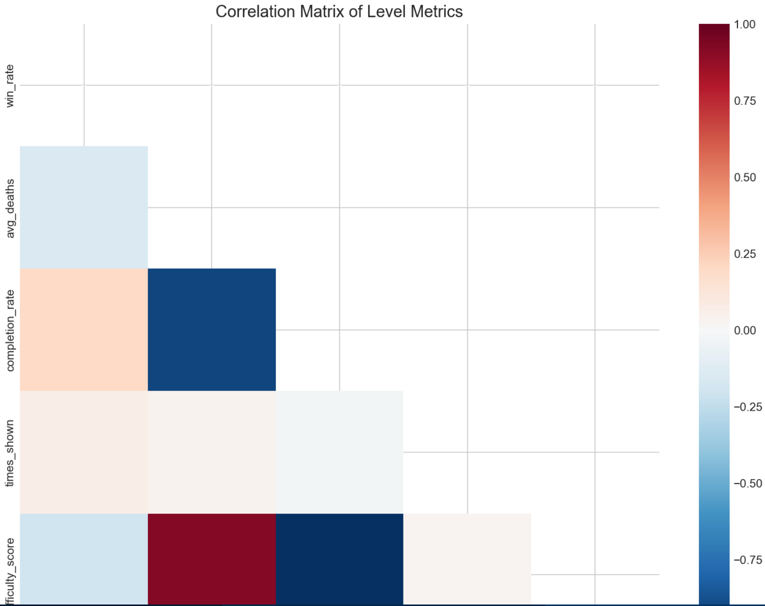


Backup: Generator Fingerprints





Backup: Correlation Matrix



Backup: Feature Importance